# SMART-G objects demo: what it computes and what it gave

Results of *demo_notebook_objects.py* from the SMART-G repository (hygeos/smartg, commit 0c04756), run on this machine (NVIDIA GeForce RTX 3090, CUDA 13.1, pycuda 2026.1, pytrunc 2.1.0). Every number and figure below was produced by that run.

## 1. What the script is about

The notebook is the **3D objects** demo of SMART-G. Its main subject is a **solar power tower**: a field of mirrors (*heliostats*) that reflect sunlight onto a small *receiver* on a tower, 106.5 m high, which here is a 12 m × 14 m panel divided into 0.5 m cells. SMART-G follows photons through the atmosphere (a 1D layered atmosphere with desert aerosol) and through these 3D objects, and reports how much power reaches each cell of the receiver. The last part is unrelated to towers: a coastal scene showing how ocean depth changes the colour of the water seen from above.
Units are kilometres inside SMART-G (the mirrors are about 9.5 m × 12.8 m, written as 0.00945 × 0.01284). All runs are at 550 nm unless said otherwise, over a ground of albedo 0.25.

### Three ways to run the same physics

| Mode | How photons are launched | Used in |
|---|---|---|
| Restricted forward (RF) | From the top of the atmosphere, aimed only at the mirrors (ray tracing). Cheap, and exact for the direct part, but photons are not sampled outside the mirror footprint. | Cases 1 and 2 |
| Full forward (FF) | From the top of the atmosphere, over a rectangle (*cfx* by *cfy*) that covers the mirrors and the surroundings. Needs many photons. | Cases 3 and 5 |
| Backward (BR) | From the receiver, towards the mirrors and the sun, then follows the light back. Efficient when the receiver is small compared to the field. | Cases 4 and 6 |

### The categories of light at the receiver

SMART-G sorts the photons that reach the receiver by what they touched on the way (the definition is in Moulana et al., 2019, cited by the code). From the labels used in the notebook: **D** = direct sunlight that hits the receiver without a mirror, **H** = light reflected by a heliostat, **E** = light scattered by the environment (the ground), **A** = light scattered by the atmosphere, and the combinations **H+A**, **H+E**, **E+A**, **H+E+A**. Category 0 is the sum. The absolute values are scaled with a top-of-atmosphere solar irradiance of 1320 W/m² (the default *mtoa* of *cat_view*), so they scale with that choice.

## 2. The runs at a glance

| Case | Setup | Photons | Kernel / total time | Result (total at the receiver) |
|---|---|---|---|---|
| 1. Quick start | 4 mirrors in a row, sun zenith 14.3°, RF, aerosol optical depth 0.25, pressure 877 hPa | 4e7 | 2.5 s / 23 s* | 2036 W/m² ± 0.011% |
| 2. Heliostats from two angles | 3 mirrors placed by angle and distance, RF, aerosol 1.0 | 1e7 | 1.8 s / 2.5 s | 2087 W/m² ± 0.043% |
| 3. Heliostats from a file | STP1.dat positions (9 mirrors), FF over 0.06 × 0.08 km, aerosol 0.2 | 4e7 | 1.9 s / 2.5 s | 4855 W/m² ± 0.036% |
| 4. Backward, sensor view | 1 mirror, rough (0.1), sun zenith 50°/azimuth 45°; a sensor at the receiver looks at the mirror | 1e8 | – | radiance map (no flux table) |
| 5. Tower with faceted mirrors, forward | 4 heliostats of 2 × 2 facets each, sun zenith 50°, FF over 8 × 8 km, aerosol 0.5 | 4e9 | 33.2 s / 34.0 s | 1605 kW ± 0.25% |
| 6. The same, backward | Same scene, receiver-side launch, sun disk 0.266° | 1e9 | 26.7 s / – | 1615 kW ± 0.22% |
| 7. Coastal depth | Land \| ocean transect, three depths, 3 wavelengths | 5e7 each (3 runs) | about 22 s in all | see section 4 |

* the first run also includes the one-time start-up of the CUDA program (about 3 s of compilation here). The percentages are SMART-G's 68% relative error estimates for the total.

## 3. The solar-tower runs

### 3.1 Quick start: four mirrors and one receiver

Four identical mirrors (reflectivity 0.88) stand every 50 m in a line, each tilted so that its reflection lands on the receiver; the receiver, 106.5 m up, takes the light. The first figure is the drawing check: the grey plates are the mirrors, the red arrows are the reflected rays pointing to the receiver on the right.

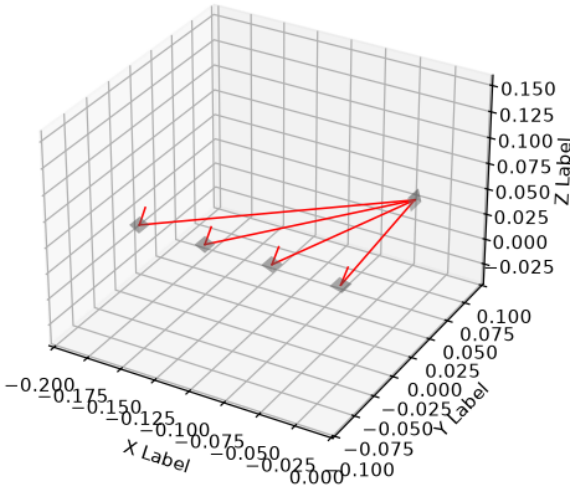



Figure 1. The drawing check of the quick-start scene (visualize_entity): four mirrors, the receiver at the right-hand end of the arrows.

The receiver gets **2036 W/m²** on average, with an estimated error of 0.011%. Almost all of it (**2026 W/m², 99.5%**) is category H, light that a mirror reflected. Category H+A (reflected and then scattered once in the air, 9.5 W/m²) is next, 0.5%. The others (A alone, E+A, H+E+A) are below 0.04 W/m² and carry large relative errors because few photons reached them (432, 1631 and 51 photons); D and E are exactly zero, as expected for a receiver that faces away from the sun and a ground that does not see it.

| Category | Irradiance (W/m²) | Photons | Error (%) |
|---|---|---|---|
| Sum (0) | 2035.84 | 27 231 204 | 0.011 |

| Category | Irradiance (W/m²) | Photons | Error (%) |
|---|---|---|---|
| H (2) | 2026.29 | 27 102 351 | 0.011 |
| A (4) | 0.037 | 432 | 4.8 |
| H+A (5) | 9.475 | 126 739 | 0.28 |
| E+A (7) | 0.035 | 1 631 | 2.5 |
| H+E+A (8) | 0.001 | 51 | 14 |
| D (1), E (3), H+E (6) | 0 | 0 | – |

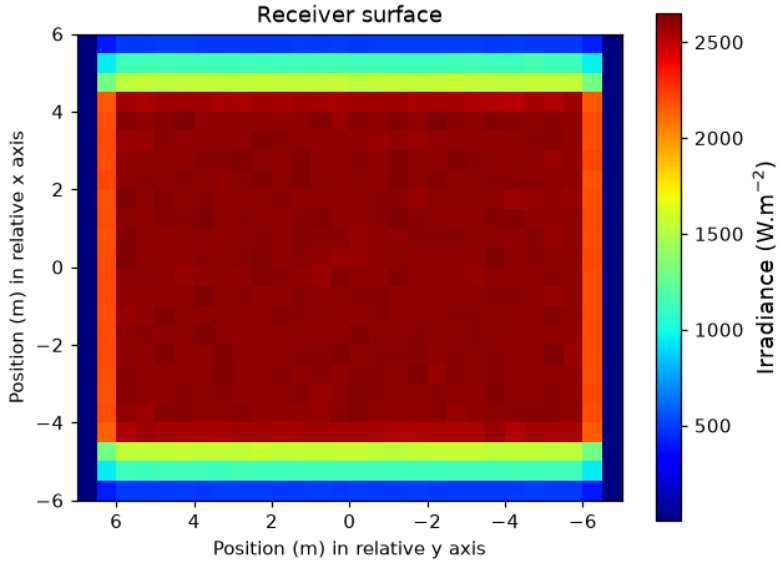



Figure 2. Irradiance over the receiver, all categories. The plateau is the part the mirrors illuminate; the rim is cut off at the edge of the cells (the receiver is 12 m × 14 m, in 0.5 m cells).

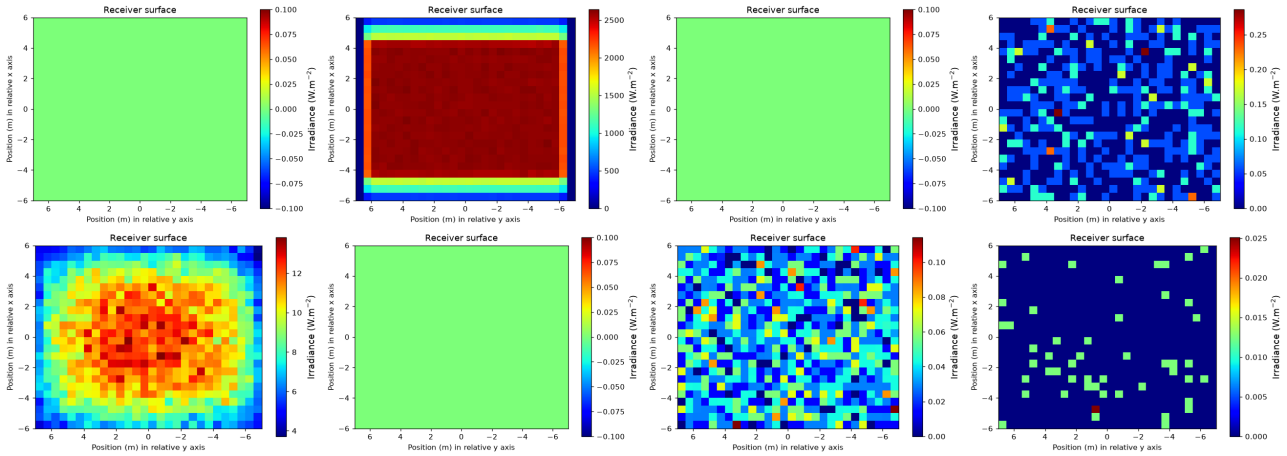


Figure 3. The receiver maps by category (note the different colour scales). Top row: D (empty), H (the flux), E (empty), A. Bottom row: H+A (smooth, 4 to 13 W/m², centred on the beam), H+E (empty), E+A, H+E+A (a handful of photons). The grainy maps are photon noise, not structure.

## 3.2 Mirrors placed by angle (case 2) and from a file (case 3)

*generate_h_a* places heliostats on a fan of angles at given distances; *generate_h_p* takes their positions from a file (here STP1.dat from SMART-G's auxiliary data), and in both cases it also tilts each one so its reflection reaches the receiver. Case 2 has three mirrors, the receiver sees **2087 W/m²** (H 2043, H+A 43.4: the aerosol optical depth is 1.0 there, four times the quick start, so the scattered share is 2.1% instead of 0.5%). Case 3 has nine mirrors (the file STP1.dat) and is run in full-forward mode: **4855 W/m²**. Here category E is no longer zero (7.2 W/m²), because photons launched over the whole rectangle can reach the receiver after a bounce on the ground (my reading; I did not trace individual photons).

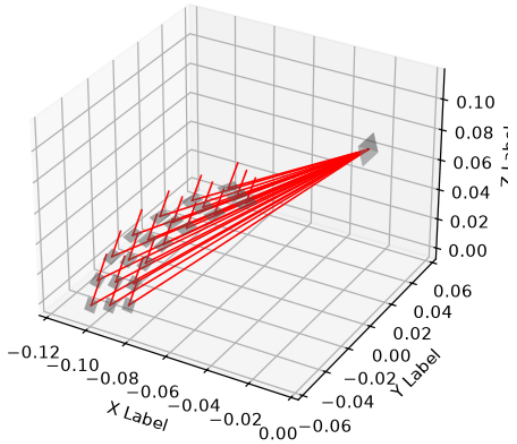



Figure 4. Case 2: three mirrors placed by two angles.

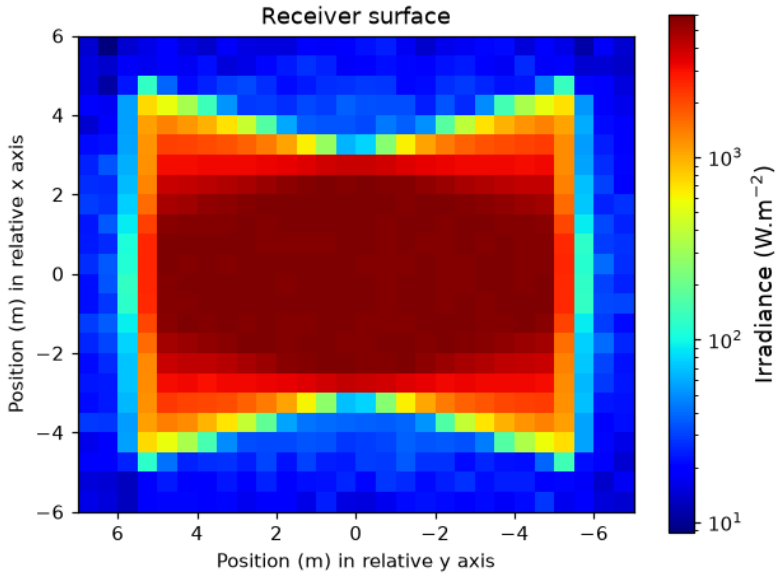



Figure 5. Case 2, receiver irradiance on a logarithmic scale: a central block of about 5000 W/m² with a halo from scattering; the dark gaps in the halo show the structure of the beam.

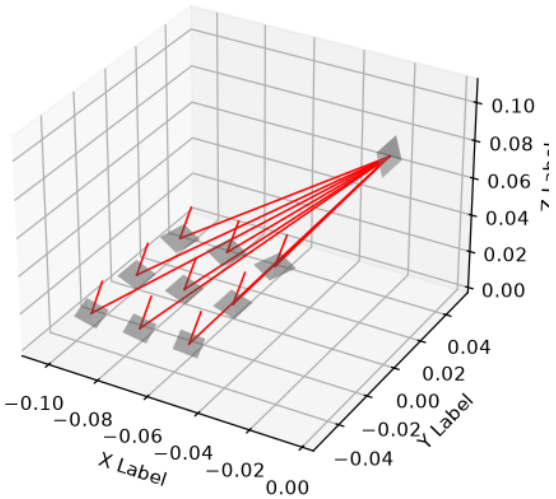



Figure 6. Case 3: the nine mirrors built from the position file (every red arrow is the reflection towards the receiver).

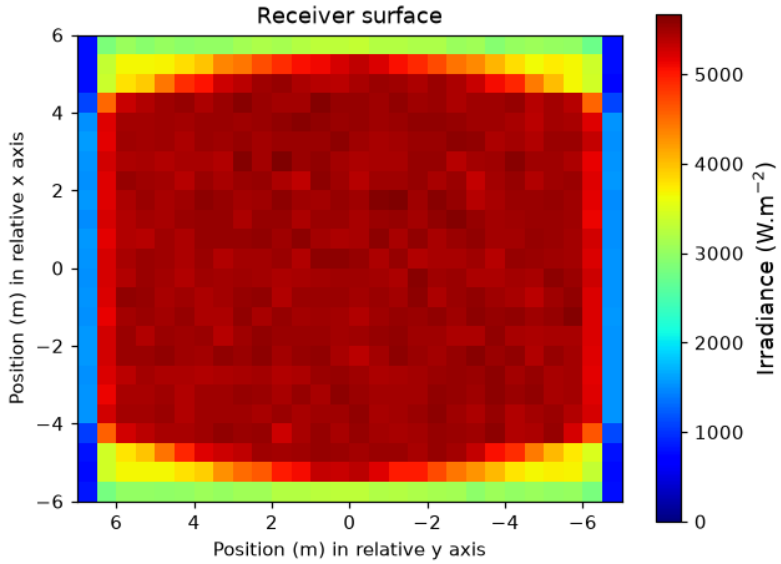



Figure 7. Case 3, receiver irradiance: a flat top at about 5500 W/m² with the same rim as before.

### 3.3 One mirror seen from the receiver (case 4, backward)

In backward mode a *Sensor* is placed at the receiver and looks at the mirror (here a single rough mirror, roughness 0.1, sun at zenith 50°, azimuth 45°). The figure is SMART-G's standard output view: polar maps of the Stokes parameters I, Q, U and the degree of polarisation, plus cuts along the zenith for several azimuths. The intensity map has a single compact bright lobe, and the cut peaks at a zenith angle of 50°, the sun's zenith angle: the mirror's glossy reflection of the sun. The polarisation components are small (Q and U within about ±0.1 against an intensity up to 26 in these units).

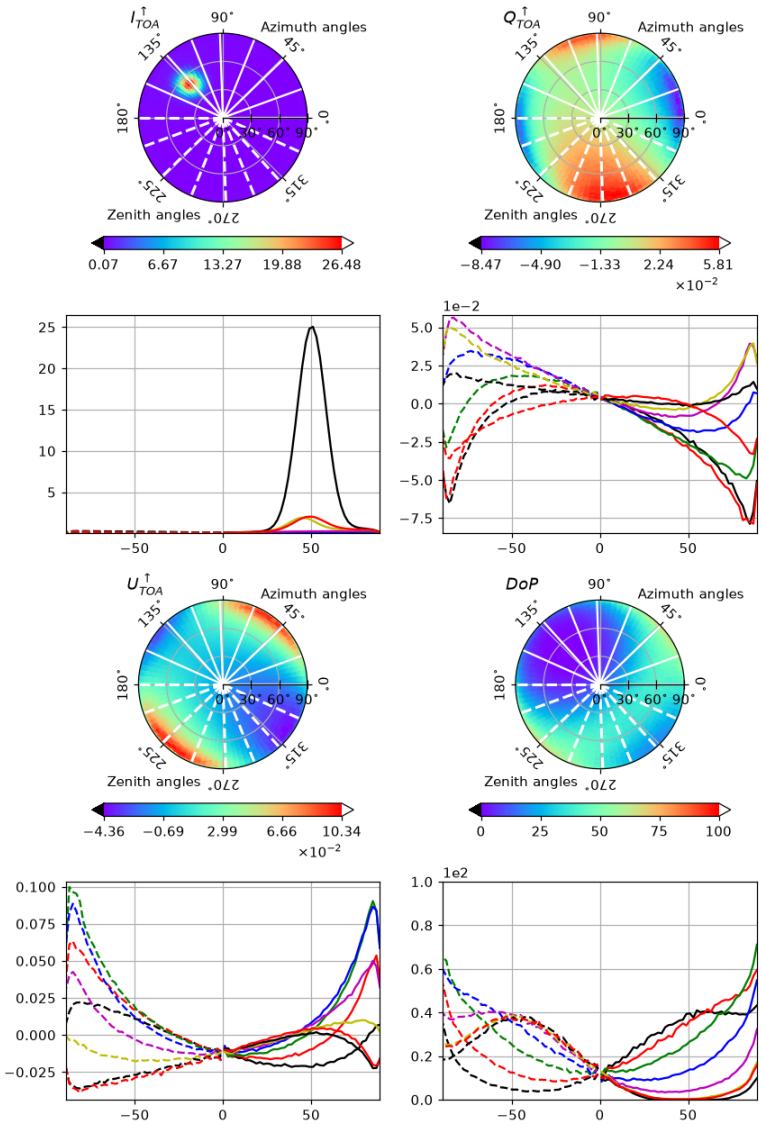



Figure 8. Case 4: radiance and polarisation seen through the rough mirror (smartg_view).

## 3.4 Faceted heliostats: forward against backward (cases 5 and 6)

This is the central test of the notebook. The same scene (four heliostats, each 9.5 m × 12.8 m and made of 2 × 2 facets, sun zenith 50°, azimuth 100°, aerosol optical depth 0.5) is simulated twice: from the sky down (full forward, 4e9 photons, 33 s of kernel time) and from the receiver outwards (backward, 1e9 photons, 27 s).

| | **Forward (case 5)** | **Backward (case 6)** |
|---|---|---|
| Total flux | 1605.1 kW ± 4.1 (0.25%) | 1614.8 kW ± 3.5 (0.22%) |
| H (via heliostat) | 1522.3 kW | 1519.0 kW |
| E (via ground) | 6.81 kW | 6.82 kW |
| A (via air) | 5.72 kW ± 4.5% | 12.57 kW ± 2.6% |
| H+A | 63.6 kW | 65.6 kW |
| E+A | 6.66 kW | 10.70 kW |
| Photons reaching the receiver | 158 281 | 212 777 |

The totals differ by 9.7 kW, **0.6%**, which is about 1.8 times the combined 1σ error (5.4 kW): consistent with the two methods describing the same physics. The dominant term H agrees to 0.2%, and E to 0.1%. The categories that involve the atmosphere (A, E+A) differ by a factor of two in a small term: forward gives 5.7 and 6.7 kW, backward 12.6 and 10.7 kW, while the sum is unaffected because it is 1% of the total. The two runs do not sample those paths in the same way (the box limiting the computation is 0.03 km high in the backward run and 0.125 km in the forward run, and the sun disk of 0.266° is set in a different way in each), and I did not check which settings account for the gap; read the A-type values as

uncertain to about a factor of two.

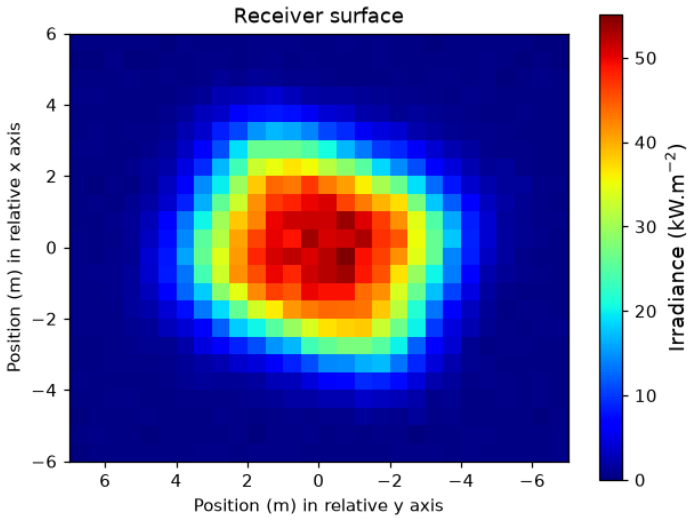



Figure 9. Receiver irradiance, forward run (case 5): a round spot peaking near 55 kW/m².

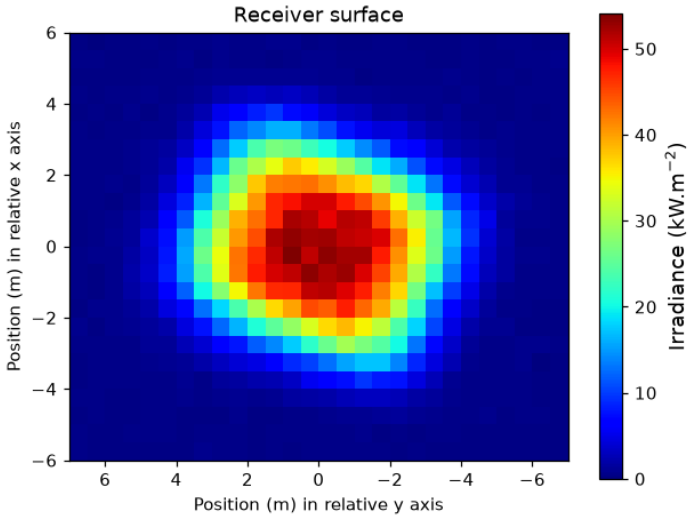



Figure 10. Receiver irradiance, backward run (case 6): the same spot, same peak. The two maps are visually the same, with the backward one slightly smoother.

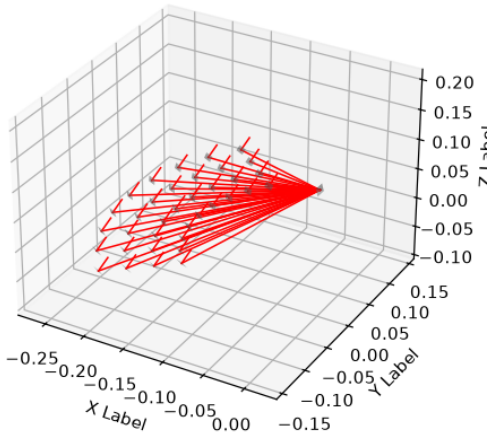



Figure 11. The drawing check of the faceted scene (shown without facets).

## 4. Coastal scene: ocean depth and colour

The last section has nothing to do with mirrors. A nadir-looking sensor at 120 km (the top of the atmosphere) scans a line from land (x < 0, albedo 0.10) to the sea (x > 0). The water is the *HydrosolPR* model with chlorophyll 0.1 mg/m³ over a sandy seafloor of albedo 0.20, and the sun is at zenith 30°. SMART-G's water is one-dimensional (one profile per run), so the author runs it three times for water 5, 20 and 100 m deep and puts the results side by side as if the depth increased offshore. The numbers are the top-of-atmosphere reflectance (I/F) at the ocean sensor nearest to 2 km offshore.

| Depth | 490 nm | 550 nm | 670 nm |
|---|---|---|---|
| 5 m | 0.146 | 0.100 | 0.036 |
| 20 m | 0.106 | 0.062 | 0.035 |
| 100 m | 0.089 | 0.057 | 0.035 |
| Land (albedo 0.10) | 0.145 | 0.121 | 0.109 |

What it shows:

- **Blue sees the bottom, red does not.** At 490 nm the signal falls from 0.146 over 5 m of water (where the bright sand still shows through) to 0.089 over 100 m, a drop of 39%. At 670 nm water absorbs the light within a few metres, so the reflectance is 0.035–0.036 at every depth: it carries no depth information.
- **5 m of water at 490 nm looks like land** (0.146 against 0.145): the sand is almost as bright as the land through the shallow layer.
- **Land is much brighter in red** (0.109 against 0.035 for the sea), while in blue the two are close when the water is shallow. The land values also rise towards blue (0.109, 0.121, 0.145) because the air itself scatters blue light more.
- **Repeatability:** across four runs the 490 nm values at 5 m were 0.1458 to 0.1470 (about 1% apart), the noise of 5e7 photons.

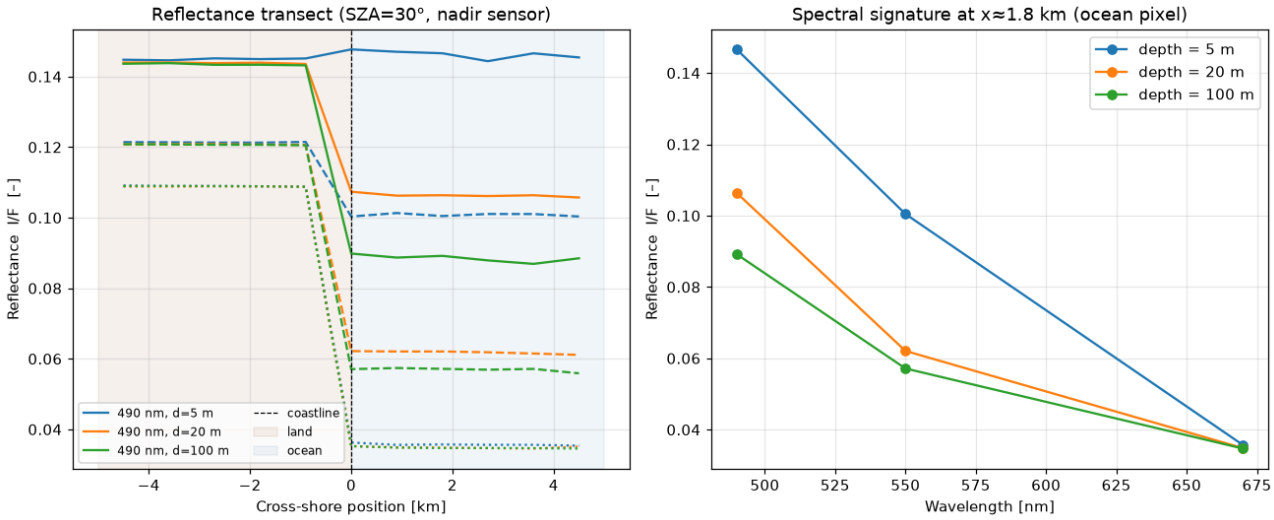



Figure 12. Left: reflectance along the transect, solid 490 nm, dashed 550 nm, dotted 670 nm, coloured by water depth. The step at the coastline (x = 0) is the land/sea change. Right: the spectrum at the ocean pixel for the three depths.

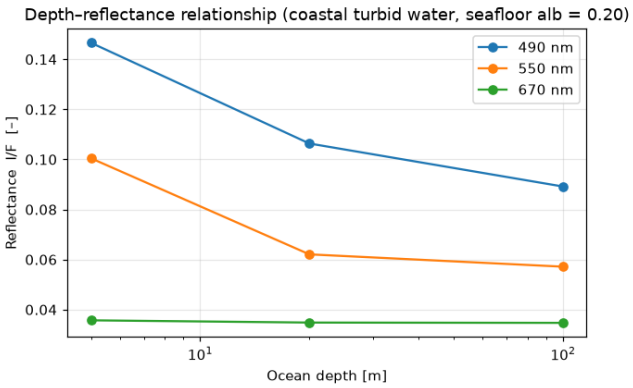



Figure 13. Reflectance against depth at the three wavelengths.

## 5. How I checked the run, and what to keep in mind

- The whole script ran to the end with exit code 0 and produced every figure above. The cells that run the 3D photon simulation use the GPU; the longest took about 34 s (case 5, 4e9 photons).
- The coastal scene uses the plain 1D engine (not the 3D objects one); like the other runs it needs a Visual Studio C++ compiler on the PATH for nvcc, as described earlier.
- The physical outputs are plausible: the mirrors deliver about 99.5% of the receiver's light with the reflectivity and the air attenuation accounted for; the forward and backward runs agree within 2σ; the water follows the expected absorption (red is absorbed first, blue last).
- **Not checked:** I did not compare against an independent reference or against the notebook's own expected values (the notebook gives none in numbers for the receiver cases). The solar constant is the default 1320 W/m² of the notebook, not a tuned value. The disagreement of the atmosphere-only categories between forward and backward (section 3.4) is unexplained.
- Random numbers: each run uses a fresh seed, so the last digits change from one run to another. The tables use one run each; the coastal numbers were also repeated four times and varied by about 1%.